IFS Projects for RDi

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Overview

The IFS Projects for RDi plug-in is used for managing RPGLE projects stored in the IFS:

- Subsystem IFS Projects in the Remote Systems view.
- Local IFS projects: projects in the RDi workspace, e.g. Git working copies, which are copied to the IFS and built there.
- Building projects and files with TOBi (The Object Builder for i).
- Building projects and files with gmake.
- SSH terminal view.
- Checking the syntax of RPGLE and SQLRPGLE files, with the errors shown in the Error List view of RDi.
- Running the build commands in the SSH terminal or in a job of the RSE connection, with the errors shown in the Error List view.

Requirements

- RDi 9.6, 9.8 or 9.9.
- The SSH server of the IBM i (`STRTCPSVR SERVER(*SSHD)`) for the terminal and for builds in the terminal.
- TOBi or GNU Make, installed from the open source packages of the IBM i.

Menu Options

The menu options are in the context menus of the *Remote Systems* view:

- *Open SSH Terminal* on an IBM i connection, *Open SSH Terminal Here* on an IFS directory, file or filter.
- *Open New SSH Terminal*, *Open New SSH Terminal Here* and *Open New SSH Terminal in Project Directory* open another terminal instead of reusing the terminal of the same directory.
- *New > IFS Project...* on the subsystem *IFS Projects* and on IFS directories.
- *Build* on a project or a file of a project.
- *Check Syntax* on an RPGLE or SQLRPGLE file of a project.
- *Open SSH Terminal in Project Directory* on a project.
- *Properties > IFS Project* on a project edits *iproj.json*.
- *Properties > IFS Project > Environment* edits the environment variables of the project.

Local IFS projects are created with the *File* menu. Their menu options are in the context menu of the project or the file, e.g. in the *Project Explorer*:

- *File > New > Project... > IFS Projects > Local IFS Project* creates a local IFS project.
- *File > Import... > IFS Projects > Existing Local IFS Project* imports a directory, e.g. a Git working copy.
- *Configure as IFS Project...* on a project with *iproj.json* makes it a local IFS project.
- *Build on IBM i* on a project or a file, *Check Syntax* on an RPGLE or SQLRPGLE file.
- *Upload Changes to IFS* and *Upload All to IFS* on a project.
- *Properties > IFS Project* edits the connection, the deploy directory and *iproj.json*.

Preferences

The preferences page is labeled as *IFS Projects*. It contains the settings of the SSH connection and of the terminals and the defaults of the IFS projects.

Restore terminals after a restart keeps the terminals, when RDi is closed. After the restart, they are shown disconnected.

Run build and compile commands selects, where Build runs: in the SSH terminal of the project (default) or in a job of the RSE connection.

The preset sets the default build and compile command for gmake or TOBi. Each control has a tool tip with a short description.

The default environment variables apply to all projects. A variable of a project overrides the default of the same name.

A more detailed description can be found on the IFS Projects project page on GitHub.

SSH keys are configured on page General > Network Connections > SSH2. JSch supports RSA and ECDSA keys in PEM format, but no Ed25519 keys.

SSH Terminal

All terminals are shown in the *SSH Terminal* view: the active terminal on the left, the list of all terminals on the right. A double-click on a disconnected terminal in the list connects it again. The context menu of the list connects, disconnects, renames and closes the selected terminals.

A terminal of the same connection and directory is reused. *Open New SSH Terminal* or the *New Terminal* button opens another one. If the view has been closed, open it with *Window* > *Show View* > *Other...* > *IFS Projects* > *SSH Terminal*.

Keys of the terminal:

- Ctrl+C copies the selected text. Without a selection, it stops the running command.
- Ctrl+V pastes the text of the clipboard.
- Shift+Page Up and Shift+Page Down scroll by one page, Ctrl+Shift+Home and Ctrl+Shift+End to the beginning or the end.
- Up, Down and Ctrl+R (command history) need bash as login shell. bash becomes your login shell with the SQL statement `CALL QSYS2.SET_PASE_SHELL_INFO('*CURRENT', '/QOpenSys/pkgs/bin/bash')`.

Syntax Check

Check Syntax compiles an RPGLE or SQLRPGLE stream file of an IFS project with option *EVENTF and shows the errors in the Error List view of RDi. A double-click on an error opens the source file or the copy book at the line of the error. For a local IFS project, the changes are copied to the IFS first.

The file is compiled in a job of its own, which is set up like the build jobs of TOBi:

- The current directory is the project directory.
- The commands of `setIBMiEnvCmd` are executed and the libraries of `curlib`, `preUsrlibl` and `postUsrlibl` are added to the library list. Variables, such as `&ENVCMD`, are replaced by the environment variables of the project.
- The directories of `includePath` are passed as include directories.

RPGLE files are compiled with CRTBNDRPG or, for NOMAIN modules, with CRTRPGMOD and option *NOGEN, so no object is created. SQLRPGLE files are compiled with CRTSQLRPGI into a module in QTEMP, because *NOGEN would skip the RPG compiler.

If the file has unsaved changes, you are asked whether to save them before the syntax check.

LPEX Editor

In the LPEX editor, the syntax check is available as LPEX user action. The plug-in does not change the LPEX profile. Add the user action on the LPEX preference pages:

1. LPEX Editor > User Actions: name `ifsCheckSyntax`, class `de.tools400.ifsprojects.lpex.CheckSyntaxAction`.
2. LPEX Editor > User Key Actions: assign a key to action `ifsCheckSyntax`.

The action can also be run on the LPEX command line:

```
action ifsCheckSyntax
```

Build in the SSH Terminal

By default, Build runs in the SSH terminal of the project, which shows the output. Afterwards the errors of the compiled objects are shown in the Error List view, as described in Build in a Job of the RSE Connection.

To learn when the build has ended, the plug-in runs the command in a subshell. The subshell writes marker files into the directory of temporary files of the plug-in (`<home>/.ifsprojects/tmp`), also when the build is interrupted with Ctrl+C or the terminal is closed. Therefore the command line in the terminal is longer than the build command. While the plug-in waits for the end of the build, a job is shown in the Progress view. Canceling it only stops waiting; interrupt the build with Ctrl+C in the terminal.

While a build is running, the next build of the project opens a new terminal. A build command with a comment (`#`) or with `&` or `|` at the end runs unchanged, without *Error List*.

Build in a Job of the RSE Connection

If Run build and compile commands is set to In a job of the RSE connection, Build runs in a job of the RSE connection, without SSH login. The output is shown line by line in the Commands Log view. Afterwards the errors of the compiled objects are shown in the Error List view. The SSH terminal remains available for interactive work.

The objects must be compiled with option *EVENTF. TOBi writes the events files into directory .evfevent of the project. For gmake builds, the events file EVFEVENT of the library specified as objlib (or curlib) in iproj.json is read.

Builds of the same project run one after the other. *Cancel* in the *Progress* view stops the build, including running compiles.

Local IFS Projects

A local IFS project is a project in the RDi workspace, e.g. a Git working copy with *iproj.json*. Before each build, the plug-in copies the changed files to a directory in the IFS, the deploy directory, and builds the project there. Files are only copied from the PC to the IFS, never back. The default deploy directory is `<home>/.ifsprojects/builds/<project>`.

Not copied are the files listed in `.deployignore` or, if it does not exist, in `.gitignore` of the project directory, and `.git`, `.project`, `.settings` and the output folders of TOBi.

If a file in the deploy directory has been changed by someone else, e.g. with Code for IBM i, you are asked whether it may be overwritten. Other files of the deploy directory, e.g. the output of TOBi, are never deleted.

The files are copied unchanged. A new stream file gets the CCSID of the encoding of the local file, e.g. 1208 for UTF-8. A double-click on an error in the *Error List* view opens the local file.

Change Log 1.0.0.r (11.10.2026)

Plug-in: 9.6.0.13+

- Initial version

IFS Projects On The Internet

Project page on GitHub	https://github.com/tools-400/ifs-terminal-plugin/
Homepage	https://tools-400.github.io/ifs-terminal/
RDi 9.6+ Update Site	https://tools-400.github.io/ifs-terminal/update-site/eclipse/rdi8.0/
RDi 9.6 Beta Update Site	https://tools-400.github.io/ifs-terminal/beta-version/update-site/eclipse/rdi8.0/